DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
UNIVERSITY OF BARISHAL
FINAL EXAMINATION
Course Title: **Mathematical Analysis for Computer Science**
Course Code: CSE-3201
3rd Year 2nd Semester
Admission Session: 2018-19
(Answer Any Five Questions)

Time: 3 Hours Marks: 60

1. a) Show that it's possible to construct a Venn diagram for all 2^n possible subsets of n given sets, using n convex polygons that are congruent to each other and rotated about a common center. [3]

   b) A smoke detector system uses two devices, A and B. If smoke is present, the probability that it will be detected by device A is .95; by device B, .90; and by both devices, 0.88. Now, [3]

   i. If smoke is present, find the probability that the smoke will be detected by either device A or B or both devices.
   ii. Find the probability that the smoke will be undetected.

   c) If W_n is the minimum number of moves needed to transfer a tower of n disks from one peg to another when there are four pegs instead of three, show that [3]

$$W_{n(n+1)/2} \leq 2W_{n(n-1)/2} + T_n, \quad \text{for } n > 0.$$

   (Here $T_n = 2^n - 1$ is the ordinary three-peg number. Use this to find a closed form $f(n)$ such that $W_{n(n+1)/2} \leq f(n)$ for all $n \geq 0$.

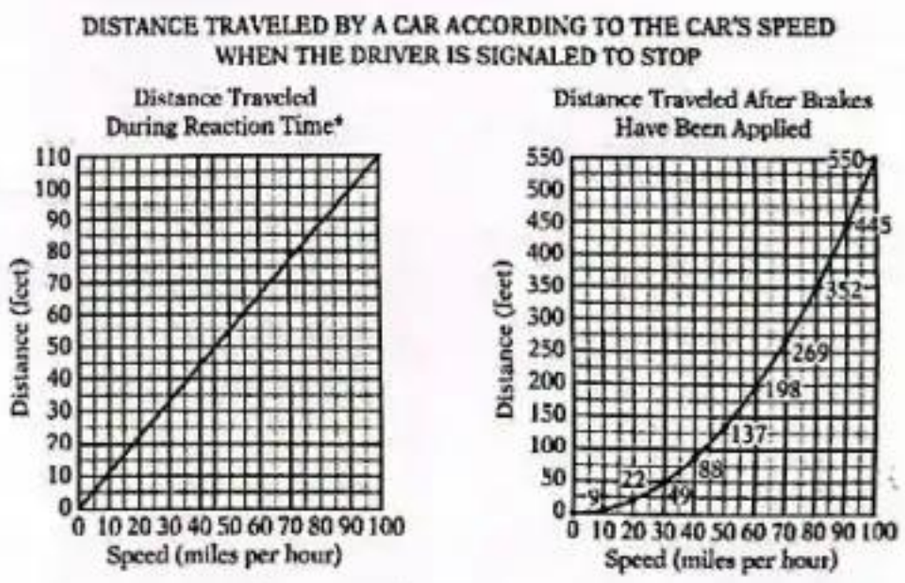
   d) Josephus had a friend who was saved by getting into the next-to-last position. What is I(n), the number of the penultimate survivor when every second person is executed? [3]

2. a) Write down the following questions with necessary explanation. [3]

A. The speed, in miles per hour, at which the car travels a distance of 52 feet during reaction time is closest to which of the following?
i. 43 ii. 47 iii. 51 iv. 55 v. 59

(B) Approximately what is the total stopping distance, in feet, if the car is traveling at a speed of 40 miles per hour when the driver is signaled to stop?
i. 130 ii. 110 iii. 90 iv. 70 v. 40

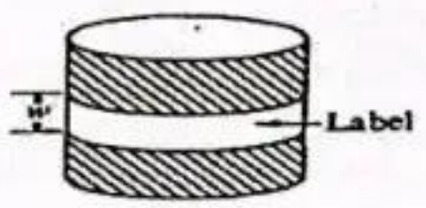
(C) Of the following, which is the greatest speed, in miles per hour, at which the car can travel and stop with a total stopping distance of less than 200 feet?
i. 50 ii. 55 iii. 60 iv. 65 v. 70



*Reaction time is the time period that begins when the driver is signaled to stop and ends when the driver applies the brakes.

Note: Total stopping distance is the sum of the distance traveled during reaction time and the distance traveled after brakes have been applied.

b) Prove that "Circumference (c) of a circle is directly proportional to its radius (r)". Establish the relationship between 'w' with 'r' [2]



A rectangular label is attached to a right circular cylinder with radius r. The label, which encircles the cylinder without overlap, has width w and an area equal to the area of the base of the cylinder.

c) A certain pet store sells only dogs and cats. In March, the store sold twice as many dogs as cats. In April, the store sold twice the number of dogs that it sold in March, and three times the number of cats that it sold in March. If the total number of pets the store sold in March and April combined was 500, how many dogs did the store sell in March? [2]

d) At a recent dog show, there were 5 finalists. One of the finalists was awarded "Best in Show" and another finalist was awarded "Honorable Mention." In how many different ways could the two awards be given out? [3]

e) What do you mean by expectation? [2]

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